A RIGHT ADJOINT OF A COCONTINUOUS FUNCTOR COMPUTES THE PUSHFORWARD AND THE PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The morphism of topoi associated to a cocontinuous functor. Same chapter, Lemma lemma-cocontinuous-morphism-topoi. Supplies the two equalities identifying the pushforward and the pullback.
- The direct image of a sheaf along a continuous functor. Same chapter, Lemma lemma-pu-sheaf. Shows that no sheafification is needed in the identification of the pushforward.
- The functors on presheaves attached to an adjoint pair. Same chapter, Lemma lemma-adjoint-functors. Identifies the functors on presheaves attached to the two functors related by adjunction.

Source reference. Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 5063–5063. Navigation pointer invoked at proof line 5063; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XK: sources/stacks-sites/pinned-original-source.tex, source lines 5084–5084. Navigation pointer invoked at proof line 5084; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 09VQ: sources/stacks-sites/pinned-original-source.tex, source lines 5085–5085. Navigation pointer invoked at proof line 5085; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 00VC: sources/stacks-sites/pinned-original-source.tex, source lines 5087–5087. Navigation pointer invoked at proof line 5087; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$, and $v : \mathcal{D} \to \mathcal{C}$ be functors. Assume that u is cocontinuous and that v is a right adjoint to u. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the morphism of topoi associated to u, see Lemma 00XO. Then*

1. *for a sheaf $\mathcal{F}$ on $\mathcal{C}$ the sheaf $g_*\mathcal{F}$ is equal to the presheaf $v^p\mathcal{F}$, in other words, $(g_*\mathcal{F})(V) = \mathcal{F}(v(V))$, and*
2. *for a sheaf $\mathcal{G}$ on $\mathcal{D}$ we have $g^{-1}\mathcal{G} = (v_p\mathcal{G})^\#$.*

Proof. For $\mathcal{F}$ as in (1) we have

$$g_*\mathcal{F} = {}_su\mathcal{F} = {}_pu\mathcal{F} = v^p\mathcal{F} = \mathcal{F} \circ v$$

The first equality is Lemma 00XO. The second equality is Lemma 00XK. The third equality is Lemma 09VQ. The final equality is the definition of v^p in Section 00VC. This proves (1). For $\mathcal{G}$ as in (2) we have

$$g^{-1}\mathcal{G} = (u^p\mathcal{G})^\# = (v_p\mathcal{G})^\#$$

The first equality is Lemma 00XO. The second equality is Lemma 09VQ. □